ComMarker Omni X Laser Safety Policy for PCB Etching

1. Purpose

This policy defines the mandatory safety rules for operating the ComMarker Omni X laser in the lab for PCB etching, marking, or material removal. It is intended to prevent injury, fire, property damage, and hazardous exposure to fumes or particulates.

Every employee, contractor, intern, visitor, or other person who operates, assists with, cleans, or observes operation of the laser must follow this policy and sign the acknowledgement before participating in laser work.

2. Scope

This policy applies to:

- Laser etching, marking, engraving, ablation, or testing on PCBs, copper-clad boards, solder mask, silkscreen, FR-4, or related electronics materials.
- Setup, focusing, framing, alignment, cleaning, inspection, maintenance, and emergency response for the laser.
- Anyone inside the laser-controlled area while the laser is powered, armed, framing, focusing, or firing.

This policy does not cover chemical PCB etching, soldering, reflow, milling, or wet chemistry, which require separate procedures.

3. Conservative Safety Assumption

The person handling the ComMarker Omni X shall treat it as a high-hazard laser system unless the manufacturer label, manual, enclosure, interlocks, and current setup show that the accessible beam is fully contained during normal operation.

If the enclosure is open, removed, damaged, bypassed, misaligned, or under service, the operator shall treat the system as a Class 4 laser hazard and shall not operate it unless the task has a specific safe procedure. Class 4 laser exposure can cause permanent eye injury, skin burns, fire, and hazardous reflections.

4. Main Ways a Person Could Be Hurt

Employees must understand that injury can occur even during short, routine jobs. The principal hazards are:

1. **Eye injury or permanent blindness.** Direct laser exposure, reflected laser exposure from copper or other shiny surfaces, and scattered radiation can injure the eye. Some laser wavelengths are invisible, so a person may not blink, feel pain, or realize an exposure occurred.
2. **Skin burns.** The beam or reflected beam can burn exposed skin. Hot PCBs, metal fixtures, and recently etched surfaces can also burn skin.
3. **Fire.** PCBs, paper, cardboard, wipes, dust, plastic, adhesives, coatings, and fixture materials can ignite. Laser fire can start quickly and may continue inside the enclosure.
4. **Toxic fumes, smoke, and particulates.** Laser interaction with FR-4, epoxy, fiberglass, solder mask, copper, tin, flux residue, adhesives, conformal coating, and unknown coatings can produce irritating or hazardous fumes and fine particulate. These may affect the lungs, eyes, skin, and throat.
5. **Electrical shock.** The laser, power supply, fume extractor, computer, and accessories can present shock hazards, especially if damaged, wet, modified, or serviced while energized.
6. **Mechanical or pinch injury.** Motorized stages, focus mechanisms, rotary fixtures, lids, clamps, and moving accessories can pinch, strike, or trap fingers.
7. **Hot, sharp, or contaminated workpieces.** PCBs may have sharp edges, fiberglass splinters, copper burrs, chemical residues, solder residues, or hot surfaces after processing.
8. **Unexpected operation.** Incorrect job files, wrong origin, wrong units, autofocus errors, software settings, reflected beams, or accidental start commands can cause unintended firing.
9. **Compressed air and exhaust hazards.** Air assist, compressed air, ducts, filters, and fans can spread smoke or debris if incorrectly configured or poorly maintained.
10. **Bypass or complacency risk.** Injuries are more likely when an operator bypasses interlocks, runs the laser unattended, uses unsuitable materials, skips PPE, or assumes a familiar job is harmless.

5. Operator Responsibilities

5.1 Person Handling the Laser

The person handling the laser shall:

- Complete required training before use.
- Follow this policy, the manufacturer's manual, and the lab operating procedure.
- Confirm the laser class, wavelength, required optical density for eyewear, enclosure condition, interlock function, emergency-stop function, exhaust function, and material suitability before use.

- Wear required PPE and require observers to do the same.
- Stay with the laser during operation.
- Stop work immediately if unsafe conditions exist.
- Report incidents, near misses, fires, smoke events, injuries, equipment faults, interlock failures, or unusual odors.
- Remove the laser from service if a safety control is missing, damaged, bypassed, unreliable, or unclear.
- Avoid any new material, process, fixture, optic, enclosure change, ventilation change, or maintenance activity unless the hazards are understood and the safe procedure is clear.

5.2 Observers and Visitors

Observers and visitors may enter the laser-controlled area only with permission from the person handling the laser. They must follow all PPE, viewing, and distance requirements. Visitors may not operate the laser.

6. Authorization and Training

No person may operate the laser independently until all of the following are complete:

- Review of this policy and signature of the acknowledgement.
- Review of the current manufacturer manual and lab operating procedure.
- Hands-on training by a person already competent with this laser and procedure.
- Demonstrated ability to inspect the laser, verify ventilation, load a PCB safely, focus/frame the job, start and stop a job, use the emergency stop, respond to smoke or fire, and clean up safely.
- Demonstrated understanding of suitable materials, prohibited materials, required PPE, and incident reporting.

Refresher review is required after any incident, near miss, equipment change, process change, or extended period without use.

7. Required Engineering Controls

The following controls must be in place before any laser operation:

- The laser enclosure, lid, viewing window, panels, and beam path must be intact and properly installed.
- Interlocks must be functional. Interlocks must not be bypassed, defeated, taped down, modified, or ignored.
- The emergency stop must be accessible and tested according to the maintenance schedule.
- The laser key, password, or access control must be limited to trained operators.

- The fume extraction system must be operating before laser firing and must continue long enough after the job to clear smoke and odor.
- Exhaust must be vented or filtered using equipment appropriate for PCB smoke and fine particulate, including filters maintained on schedule.
- The workpiece must be secured on a stable, non-reflective, fire-resistant fixture.
- Reflective or shiny materials near the beam path must be removed, covered, or controlled.
- A suitable fire extinguisher must be visible, accessible, inspected, and appropriate for the likely fire class in the lab.
- Warning signage must be posted at the laser area when the laser is in use or armed.

If any required control is missing or defective, the laser must not be operated.

8. Required PPE

Minimum PPE for normal enclosed operation:

- Safety glasses.
- Gloves for handling PCBs, fixtures, debris, or recently processed material.

Additional PPE is required whenever the enclosure is open, interlocks are disabled for service, alignment is performed, or the task creates a credible exposure risk:

- Laser safety eyewear rated for the exact laser wavelength and adequate optical density for the maximum credible exposure.
- Non-reflective tools and fixtures.
- Skin protection appropriate to the task.
- Any respiratory protection only if selected and managed under the lab's respiratory protection program. Disposable dust masks are not a substitute for proper exhaust.

Operators must not wear watches, rings, bracelets, reflective jewelry, or other reflective items that could enter the beam area.

9. Pre-Use Checklist

Before each job, the operator shall:

1. Confirm they are trained and not impaired, distracted, or rushed.
2. Confirm the material and process are suitable for laser processing.
3. Inspect the enclosure, viewing window, interlocks, lid, emergency stop, cables, power supply, lens, work bed, fixtures, and exhaust.
4. Remove paper, wipes, solvents, aerosols, clutter, loose parts, and unnecessary combustibles from the laser area.
5. Confirm the fume extractor is on, airflow is present, ducts are connected, and filters are not expired or overloaded.

6. Confirm the fire extinguisher is accessible.
7. Secure the PCB flat and stable. Do not hold a workpiece by hand during operation.
8. Confirm the job file, material settings, units, origin, focus, framing, and expected laser path.
9. Use the lowest practical power and shortest practical exposure for the task.
10. Confirm observers are outside the controlled area or wearing required PPE.
11. Close the enclosure and confirm interlock status before firing.

If anything is uncertain, stop and do not operate the laser until the safe setup and procedure are clear.

10. Operating Rules

Operators shall follow these rules during every job:

- Do not run the laser unattended. Remain close enough to stop the job immediately.
- Watch the start of every job and monitor for smoke, flame, shifting material, unexpected reflection, wrong origin, wrong scale, or unusual noise.
- Do not open the enclosure while the laser is firing.
- Do not reach into the enclosure until the laser has stopped, the job is complete, fumes have cleared, and the workpiece has cooled.
- Do not bypass, override, defeat, or modify safety features.
- Do not operate with cracked windows, missing panels, damaged seals, exposed wiring, abnormal fan noise, poor exhaust, or visible smoke leakage.
- Do not leave the laser powered and armed when not actively setting up or running a job.
- Do not process unsuitable materials or test unknown scraps.
- Do not use a phone, headphones, or unrelated computer work in a way that prevents active monitoring.
- Stop immediately if the job behaves differently than expected.

11. Alignment, Focusing, Service, and Maintenance

Routine focusing and framing must be performed using the manufacturer's intended safe method.

Any open-enclosure alignment, service, repair, optical adjustment, interlock testing, firmware modification, electrical troubleshooting, or operation with guards removed may be performed only by a person competent for that task under a specific safe procedure.

During service:

- Use lockout/tagout or equivalent power isolation when required.
- Use the lowest possible power and shortest possible exposure.
- Use beam blocks and non-reflective tools.
- Keep eyes and skin out of the beam plane.

- Require all people in the controlled area to wear wavelength-appropriate laser eyewear.
- Restore all guards, covers, interlocks, exhaust, and access controls before returning the laser to service.

12. Post-Use Procedure

After each job, the operator shall:

1. Allow exhaust to run until smoke and odor are cleared.
2. Confirm the laser is no longer firing and is safe to open.
3. Allow the PCB and fixture to cool before handling.
4. Remove the workpiece using gloves or tools if sharp, hot, dirty, or contaminated.
5. Clean debris only after the laser is off or in a safe state. Do not blow dust into the room.
6. Dispose of PCB debris, wipes, and filters according to lab waste procedures.
7. Turn off or secure the laser so untrained users cannot operate it.

13. Emergency Procedures

13.1 Suspected Eye or Skin Exposure

- Stop the laser immediately using the emergency stop.
- Do not continue the job.
- Stop work and notify the person responsible for the lab or equipment immediately.
- Seek medical evaluation urgently for suspected eye exposure, even if there is no pain or visible injury.
- Preserve the job file, settings, material, and equipment state for investigation.

13.2 Fire

- Press the emergency stop.
- If safe, stop the job and disconnect power according to the lab emergency procedure.
- Use the fire extinguisher only if trained, the fire is small, and there is a clear exit path.
- Evacuate and call emergency services if the fire grows, produces heavy smoke, involves electrical equipment, or cannot be controlled immediately.
- Do not resume operation until the equipment has been inspected and the cause has been corrected.

13.3 Heavy Smoke, Strong Odor, or Ventilation Failure

- Stop the laser immediately.
- Keep the enclosure closed if safe to contain fumes.
- Leave the area if fumes escape into the room.
- Notify the person responsible for the lab or equipment.

- Do not resume until ventilation is restored and the cause is understood.

13.4 Electrical Fault

- Stop work.
- Do not touch exposed wiring, damaged cords, wet equipment, or energized parts.
- Disconnect power only if safe to do so.
- Tag the machine out of service and notify the person responsible for the lab or equipment.

14. Stop-Work Authority

Every employee has authority to stop laser work if they believe conditions are unsafe. Work may resume only when the unsafe condition is corrected and the person handling the laser has confirmed the setup is safe.

15. References for Safety Basis

- OSHA Technical Manual, Section III, Chapter 6, "Laser Hazards."
- ANSI Z136 series, especially ANSI Z136.1, "Safe Use of Lasers."
- Manufacturer manual, label, and specifications for the exact ComMarker Omni X unit installed in the lab.
- Safety data sheets and supplier information for PCB substrates, solder mask, coatings, adhesives, and cleaning chemicals used in the lab.

16. Acknowledgement

By signing below, I acknowledge that:

- I have read and understood this ComMarker Omni X Laser Safety Policy.
- I understand the ways the laser can injure me or others, including eye injury, burns, fire, fumes, electrical hazards, mechanical hazards, and hot or sharp workpieces.
- I agree to operate the laser only when trained, unimpaired, and following the safe procedure.
- I agree not to bypass safety controls, operate the laser unattended, use unsuitable materials, or allow untrained operation.
- I agree to report incidents, near misses, equipment faults, and unsafe conditions immediately.

Signature	Date